

Week 3 Summary – NumPy and Pandas for Data Manipulation

Objective

The objective of Week 3 was to learn the fundamentals of **NumPy** and **Pandas** for data manipulation and analysis. The focus was on performing numerical operations, organizing data into DataFrames, cleaning datasets, and generating meaningful insights.

Concepts Learned

- Understood the basics of **NumPy** and its advantages for numerical computing.
- Learned how to create and manipulate **NumPy arrays**.
- Performed array operations such as addition, subtraction, multiplication, and division.
- Learned the basics of **Pandas Series** and **DataFrames**.
- Created DataFrames from dictionaries.
- Accessed rows and columns in a DataFrame.
- Used `head()` to view records.
- Handled missing values using `dropna()`.
- Calculated statistical measures such as **average, maximum, and minimum**.
- Used `groupby()` to summarize data based on categories.

Hands-on Programs Completed

1. NumPy array operations
2. Pandas DataFrame creation
3. Calculating the average marks of students
4. Removing missing values from a DataFrame
5. Grouping students by class and calculating average marks

Client Project

Project Title: Student Marks Data Analyzer

Project Description

Developed a Python program using **NumPy** and **Pandas** to analyze student marks. The program creates a DataFrame, removes missing values, calculates the average, highest, and lowest marks, and groups students by class to find the average marks for each class.

Skills Used

- NumPy Arrays
- Pandas DataFrames
- Data Cleaning (`dropna()`)
- Data Aggregation (`groupby()`)

- Statistical Analysis (`mean()`, `max()`, `min()`)
- Data Processing

Learning Outcome

This week strengthened my understanding of data manipulation using NumPy and Pandas. I learned how to organize, clean, analyze, and summarize datasets efficiently. These skills are essential for Data Science and provide a strong foundation for working with real-world data.